MIL-M-0036314A (DSA-DM)
9 January 1973
USED IN LIEU OF
MIL-M-36314 (DM)
28 October 1964

MILITARY SPECIFICATION

MICROTOME, Rotary, Automatic

This limited coordination Military Specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but has not been approved for promulgation as a coordinated revision of Military Specification MIL-M-36314 (DM). It is subject to modification. However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers one type of automatic-feed, rotary microtome, suitable for serial sectioning of specimens.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

FSC 6640

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Specifications (cont'd)

Military

MIL-M-7298	Manual, Technical: Commercial Equipment.
MIL-K-36311	Knives, Microtome.
MIL-O-36312	Object Disks, Microtome.

STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

3. REQUIREMENTS

3.1 Material. Material shall be suitable for the purpose intended and as herein specified.

3.2 Construction. Microtome shall be the rotary type, 16 ± 1 inches long, 14 ± 1 inches wide, 8-3/4 ± 1/2 inches high, and shall consist of a base, object drive, feed mechanism, knife holder and cover. Except for the terminal components (drive wheel and slide block), the horizontal feed mechanism shall be entirely independent of the vertical movement of the object drive and shall be built into the base casting. Weight of microtome (without accessories) shall be not less than 72 pounds.

3.2.1 Base. Base shall be gray cast iron and shall furnish support for the knife holder, object drive and feed mechanism.

3.2.2 Object drive. Object drive shall be direct and shall consist of a hand (drive) wheel, shaft and slide block. Wheel

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shall be mounted on the right side of the microtome (knife holder proximal to the user). Movement of the slide block shall be vertical. Each clockwise rotation of the drive wheel shall cause one excursion of the slide block.

3.2.3 Feed mechanism. Feed mechanism shall enable the microtome to cut specimens of desired thickness by correlating movement of the hand wheel and slide block with the indicator (3.2.4) and shall consist of a drive wheel, eccentric cam, feed panel, ratchet, connecting rods and slide block. Clockwise rotation of the hand wheel shall move the slide block forward toward the knife holder. Ratchet wheel shall be the connecting point between the indicator and feed mechanism. Ratchet wheel shall be notched with each notch representing 1 micron of forward movement of the slide block. Feed panel shall disengage from the ratchet wheel when the hand wheel is turned in a counterclockwise direction or the slide block has moved forward 28 millimeters (mm.). Means shall be provided to reverse the feed mechanism so that the slide block may be moved to its original position.

3.2.4 Indicator. Indicator shall consist of a conveniently placed knob, micrometer screw and an easily visible scale. Knob shall turn the micrometer screw and the scale. Gearing between the micrometer screw and the ratchet wheel shall be such that specimens may be cut at any desired thickness from 1 to 50 microns. Scale shall be permanently marked from 0 to 50 in 1 micron intervals; zero and each succeeding fifth interval shall be numbered.

3.2.5 Knife holder. Major portions of the knife holder shall be fabricated of corrosion-resisting steel and shall furnish rigid stress for clamping of the knife. There shall be two clamps, each clamp at least 1/2 inch wide and so placed that they do not interfere with the cutting of specimens. The holder shall contain devices which make it possible to adjust the position of the knife while the knife is secured in the holder. Forward, backward, lateral, vertical and tilt adjustments shall be possible. Each adjustment shall be independent and the position of the knife edge shall not change, other than in the direction of the adjustment being made. Reference scale which is marked to indicate knife tilt in 2-degree intervals shall be placed near the holder. Range of tilt adjustment and graduated portion of reference scale shall be at least 30 degrees.

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3.3 Accessories. The following accessories shall be furnished with the microtome.

a. Two knives. Knives shall be in accordance with MIL-K-36311 (DM), size 2 except that one knife may be furnished without a handle (see 6.4).

b. Three object disks. Object disks shall be in accordance with MIL-O-36312 (DM), sizes 1, 2 and 3 (one each) (see 6.4).

c. One safety razor blade holder. Holder shall be fabricated of corrosion-resisting steel and shall be 4.625 ± 0.010 inches long. Holder shall have a clamp 1.4375 ± 0.002 inches long which is centered lengthwise between the ends of the holder. Clamp shall be capable of securely locking an extra heavy or wafer thin razor blade in place and shall provide even pressure throughout the length of the razor blade.

d. Three packages of razor blades (10 each). Razor blades shall be extra heavy steel, single-edge razor blades. Blades shall be 0.009 ± 0.0005 inch thick, 1-9/16 ± 5/64 inch long, and 23/32 ± 5/64 inch wide. Hardness, measured at the edge of the blade adjacent to the cutting edge, shall not be less than 795 and not more than 857, using a pyramidal diamond and a 5 kilogram (kg.) load on the Vickers hardness tester. Steel used in the manufacture of the blades shall have a carbon content of not less than 1.15 percent and not greater than 1.35 percent, and shall show a uniform, well-spheroidized cementite structure, with carbide particles of uniform size and distribution, free of laminations and non-metallic inclusions. The surface shall be free of rust and scale.

e. One 3-ounce can of pike oil.

3.4 Performance.

3.4.1 Skipping. The ratchet wheel shall not move when tested as indicated in 4.4.1.1, shall move one tooth when tested as specified in 4.4.1.2, and shall make one complete revolution, plus one tooth, minus zero teeth, when tested as specified in 4.4.1.3.

3.4.2 Thickness accuracy. The slide block shall move forward 50 ± 1 microns when tested as specified in 4.4.2.

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3.5 Finish. Cover shall be finished with gray epoxy resin.

3.6 Identification marking. Each microtome shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.7 Service data. The contractor shall furnish with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298 Manual, Technical, Commercial Equipment. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.8 Workmanship. Workmanship shall be first class throughout. Microtomes shall be free from defects listed in Table III and other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified

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herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the razor blades, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the component, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I. Unit of product shall be one microtome.

Table I For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. Unit of product shall be one microtome. Acceptance number shall be zero.

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Table II For testing

Characteristic	Requirement	Test procedure	Inspection level
Skipping	3.4.1	4.4.1.1 4.4.1.2 4.4.1.3	100% of microtomes shall be tested.
Thickness accuracy	3.4.2	4.4.2	S-1

4.3 Examination. The microtomes shall be examined for defects including, but not limited to, those listed in Table III.

Table III Classification of defects

Category	Defect
Major	
101	Component part missing.
102	Service data missing.
103	Microtome not free of sharp edges (excluding knives and razor blades).
104	Microtome not free of rust spots.
105	Applicable components not secured.
106	Indicator markings not complete, correct, legible or permanent.
Minor	
201	Painted surfaces not free of blisters and cracks.
202	Microtome not free of scratches.
203	Identification marking not complete, correct, legible or permanent.

4.3.1 Dimensional examination. The microtomes shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Skipping. The following tests shall be performed.

4.4.1.1 Zero micro setting. Set indicator at zero and rotate hand wheel.

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4.4.1.2 One micron setting. Set indicator at 1 micron and rotate hand wheel on revolution.

4.4.1.3 Complete revolution. Divide the number of teeth on the ratchet wheel by 10. Set quotient on the indicator. Rotate the hand wheel 10 revolutions at three revolutions per minute.

4.4.2 Thickness accuracy. A precision dial gage of known accuracy and graduated in microns shall be secured to the knife holder. Indicator shall be set at 10 microns and the hand wheel shall be rotated five revolutions. Object disk may be secured to the slide block. Any tolerance on the dial gage shall be subtracted from the tolerance specified in 3.4.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One microtome, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit preparation. Accessories shall be unit packaged Level A in accordance with unit package requirements as indicated in the applicable specification for the particular item. Knife holder shall be packaged in a carton of appropriate size and design. Each razor blade shall be placed in an individual paper envelope. Ten blades shall be placed in a carton of appropriate size and design. All accessories, including the 3-ounce can of pike oil, shall be overpacked in a carton of appropriate size and design, and identified as to contents and quantity of each accessory.

5.1.2.2 Unit package. Each unit, with component parts, and accessories packaged as stated above, shall be packaged in a carton of appropriate size, constructed in accordance with PPP-B-636, type CF, class weather-resistant, V3c. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents.

5.1.2.3 Closure. Closure shall be as specified in the appendix, and waterproofing shall be in accordance with 30.4 of PPP-B-636.

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5.1.3 Level C. Units shall be packaged in standard commercial containers of the type, size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Microtome is intended for use in cutting thin specimens for microscopic examination.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Federal Stock Number (FSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification covers the following item appearing in the Federal Supply Catalog:

Federal Stock Number	Item Identification
6640-433-6000	MICROTOME, Rotary, Automatic.

6.4 The knives referred to in 3.3 a are FSN 6640-433-9500, Knife, Microtome, 120 mm. The object disks referred to in 3.3 b are:

Federal Stock Number	Item Identification
	OBJECT DISK, Microtome
6640-434-1050	7/8-in. Diameter
6640-434-1060	1-1/8-in. Diameter
6640-434-1070	1-1/2-in. Diameter

6.5 This specification does not cover all types, classes grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0768

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

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